

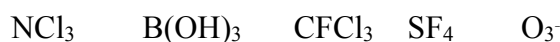
Answer ALL five questions in this section

1. Answer all parts (a) - (c).

- (a) What are the Pauli Exclusion Principle and Hund's Rule of Maximum Multiplicity? How do these determine the electronic configurations of elements? [4]
- (b) Explain what is meant by the terms *shielding* and *effective nuclear charge* as applied to atoms. [4]
- (c) Account for the variation in first ionization energies (eV) of the following Group 2 elements. [2]

Be	Mg	Ca	Sr	Ba
9.32	7.64	6.11	5.69	5.21

2. Describe the shape of *THREE* of the following species, in each case sketching the arrangement of atoms, depicting the symmetry elements and stating the point group. [10]



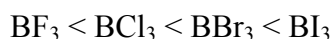
3. Answer all parts (a) to (c).

- (a) Give the Brønsted-Lowry and Lewis definitions of the terms 'acid' and 'base'. [2]

- (b) Rationalize the first pK_a values of the following species:



- (c) Explain why the Lewis acidity of boron trihalides follows the trend:



but the trend for silicon tetrahalides is:

